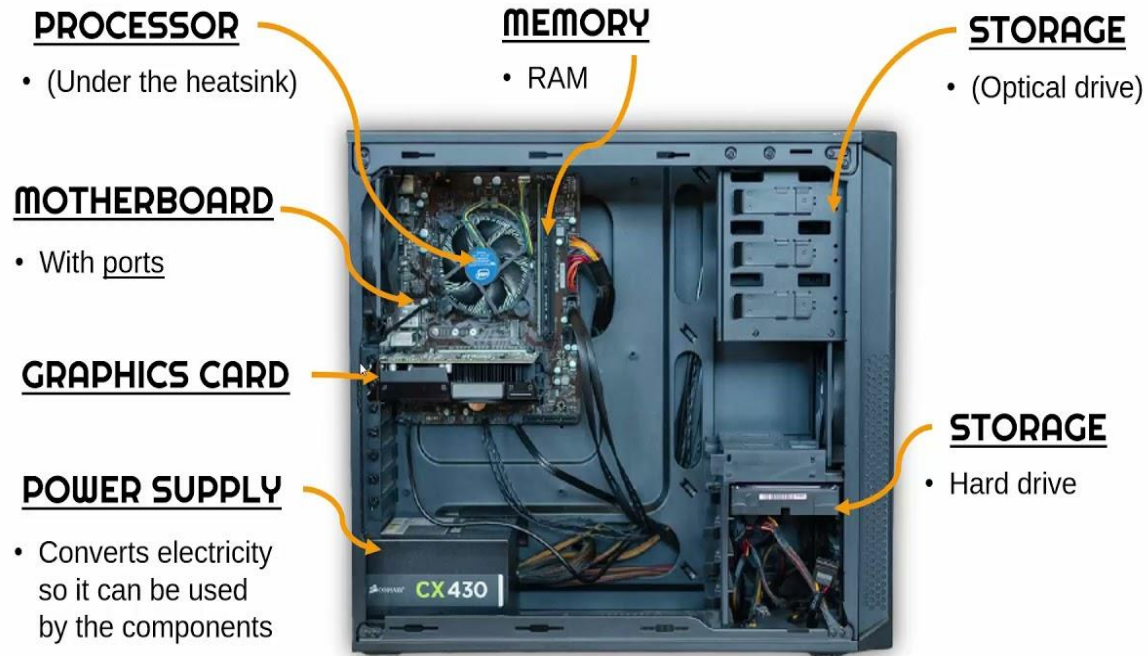


DAY-1

1.1 Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



DAY-1

1.2 Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Storage Technology	Uses spinning magnetic disks	Uses flash memory (no moving parts)
Speed	Slower read/write speeds	Much faster read/write speeds
Boot Time	30–60 seconds (typical)	10–20 seconds (typical)
Durability	Less durable; moving parts can be damaged by shocks	More durable; shock-resistant because it has no moving parts
Noise	Produces noise while operating	Silent operation
Power Consumption	Uses more power	Uses less power
Heat Generation	Generates more heat	Generates less heat
Cost per GB	Cheaper	More expensive
Storage Capacity	Available in very large capacities at lower cost	Usually smaller capacities for the same price
Typical Use Cases	File storage, backups, budget desktops, CCTV systems	Laptops, gaming PCs, high-performance desktops, operating systems, and frequently used applications

DAY-1

1.3 Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU (Central Processing Unit):

The CPU acts as the brain of the computer. In games like PUBG or Free Fire, it processes game logic, player actions, enemy AI, and physics, ensuring the game runs smoothly.

RAM (Random Access Memory)

RAM temporarily stores the game data that the CPU and GPU need while the game is running. More RAM helps reduce loading times and allows smoother gameplay, especially when multitasking.

GPU (Graphics Processing Unit)

The GPU is responsible for rendering graphics, including characters, environments, lighting, and visual effects. A powerful GPU delivers higher frame rates, better image quality, and a smoother gaming experience.

DAY-1

1.4 Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

GPU (Graphics Card):

A better GPU improves graphics quality, increases frame rates (FPS), and provides smoother gameplay at higher resolutions.

CPU (Processor):

A faster CPU handles game calculations, AI, and physics more efficiently, reducing lag and improving overall performance.

RAM:

Upgrading to **16 GB or more** allows the game and background applications to run smoothly without slowing down the system.

SSD (Solid State Drive):

Installing the game on an SSD reduces boot times, speeds up game loading, and shortens loading screens.

Cooling System:

Better air or liquid cooling keeps the CPU and GPU at safe temperatures, preventing overheating and performance drops during long gaming sessions.